

Evaluating the 3-Layer Transformer Encoder for Pakistan Sign Language Recognition

Introduction

Developing a robust mobile application for sign language translation requires an architecture capable of understanding complex spatial and temporal sequences. For our initial approach to recognizing Pakistan Sign Language (PSL), we implemented a 3-Layer Transformer Encoder. While Transformers have revolutionized natural language processing, adapting them to the visual, continuous, and low-resource domain of PSL presented a unique set of theoretical advantages and practical hurdles. This section details the architecture, the initial successes, and the critical generalization failures that ultimately led to our decision to pivot to a different model.

Why We Chose This Model

The decision to utilize a Transformer Encoder was heavily inspired by foundational research in continuous sign language recognition, notably the work by Camgoz et al. Sign language is inherently sequential; the meaning of a gesture relies not just on a static hand shape, but on the trajectory, speed, and relationship between the hands and the body over time.

Traditional models like LSTMs or RNNs process frames sequentially, which can lead to a bottleneck in learning long-range dependencies. Transformers, however, utilize a Self-Attention mechanism. This allows the model to look at an entire sequence of frames simultaneously, assigning "attention" to the most critical parts of the movement regardless of when they occur in the sequence. We hypothesized that this global context awareness would make it highly effective at distinguishing subtle differences in PSL vocabulary.

Architecture Overview

Our pipeline was designed to be lightweight enough for a mobile application while maintaining deep analytical capabilities.

- **Feature Extraction:** Instead of feeding raw video pixels into the model, we utilized MediaPipe Holistic to extract 75 3D keypoints per frame (33 pose landmarks and 21 for each hand). This reduced the computational load and focused the model purely on human movement.
- **Positional Embedding:** Because Transformers process all frames at once, they inherently lack a sense of time. We injected positional embeddings into the keypoint data to retain the chronological order of the gesture.
- **The 3-Layer Encoder:** The core of the model consisted of three stacked Transformer Encoder blocks. Each block utilized Multi-Head Attention (with 4 heads) to find spatial-temporal relationships, followed by Layer Normalization and a Feed-Forward Network to process the learned representations.
- **Classification Head:** A Global Average Pooling layer compressed the temporal sequence into a single dense vector, which was then passed through a final Softmax classification layer to predict the specific sign.

Has This Been Used in PSL Before?

Pakistan Sign Language is a notoriously low-resource language in the machine learning space. While 3D keypoint extraction and Transformer models have been heavily researched for American Sign Language (ASL) and German Sign Language (DGS), applying a purely skeletal 3-Layer Transformer Encoder to a custom PSL dataset is highly novel. Most existing PSL projects rely on simpler Image Classification (CNNs) on static frames or shallow LSTMs. Our attempt pushed the boundary of sequence modeling for PSL, but it also highlighted why cutting-edge NLP architectures struggle without massive, diverse datasets.

Benefits and Initial Success (The "Seen" Data)

During the initial training phases, the model demonstrated exceptional learning capacity. The results on the testing split of the *seen* data were remarkable:

- **Accuracy:** The model achieved a 97.82% overall accuracy (an error rate of just 2.18%).
- **Precision and Recall:** Across 413 validation samples, the macro and weighted averages for precision, recall, and F1-score all hit 0.98.
- **Class Mastery:** For complex chronological signs like "afternoon," "friday," "calendar," and "century," the model achieved perfect 1.00 precision and recall scores. It learned the exact skeletal sequences provided in the training set with near-perfect fidelity.

Challenges: The Overfitting Dilemma & Our Decision

Despite the flawless metrics on the training data, the model exhibited a catastrophic failure when deployed in real-world, unseen conditions. When tested on unseen signers or slightly varied environments, the accuracy plummeted to an unusable 30-40%.

This exposed a massive challenge: **Severe Overfitting**. The 3-layer Transformer is an incredibly "deep" and parameter-heavy architecture. We established a baseline test using a subset of 55 distinct signs, but given the relatively small size of our augmented dataset for these classes, the model did not learn the generalized concept of a PSL sign. Instead, it memorized the specific micro-movements, body proportions, and exact speeds of the specific signers in the training data. Even with dropout layers and early stopping implemented, the Transformer's high capacity meant it simply mapped exact inputs to exact outputs, acting more like a lookup table than a generalized AI.

Because of this stark contrast between high training accuracy and poor real-world generalization during the 55-sign test, we made the definitive decision not to move forward with the Transformer architecture for our final application. The ultimate goal of our Final Year Project is to create an accessible, functional mobile application for the Deaf and hard-of-hearing community. A translation app that only works for the person who trained it is fundamentally useless.

The Transformer was a highly successful proof-of-concept for sequence memorization. However, instead of fighting the architecture, we pivoted to explore models better suited for smaller datasets, such as Graph Convolutional Networks (GCNs). These models enforce strict

spatial relationships and prevent the network from memorizing arbitrary sequence noise, leading us closer to true language generalization.